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IS-NPMD-DERMATOLOGY

Soft Tissue Infections-Wounds And Burns

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TUCOM-CA

LEARNING & TESTABLE OBJECTIVES

1. List the major pathogens of burns infections and wound infections

For each of the pathogens listed in the bug table,

2. Describe the morphology, cultural and identification characteristics
3. Describe the relevant epidemiology including reservoir/ source and mode of transmission
4. Associate it with disease presentation and possible complications
5. Correlate the virulence factors with the pathogenesis of the disease
6. Describe the laboratory diagnosis
7. Select the antimicrobial of choice and describe its mode of action
8. Choose the appropriate preventive/ prophylactic measures

Pathogen	Disease
Pseudomonas aeruginosa *Staph aureus and S. epidermidis*	**Burns infections**
Clostridium perfringens and other clostridia	Gas gangrene/ Myonecrosis
Streptococcus pyogenes	Necrotizing fasciitis
Non sporing anaerobes: Bacteroides, Prevotella, Fusobacterium, Peptococcus, Peptostreptococcus	Deep seated infections
Vibrio vulnificus sea water	Cellulitis and abscess
Erysipelothrix rhusiopathiae	Seal finger/ Whale finger/ Erysipeloid
Eikenella	Human–bite associated infection
Capnocytophaga	Dog-bite associated infection
Pasteurella multocida	Cat bite/ dog bite associated infection
Bartonella henselae	Cat Scratch disease

MAJOR PATHOGENS

Traumatic Wounds

- *P. aeruginosa*
- *Clostridium* spp
- *S. pyogenes*
- Sea water – *Vibrio vulnificus*
- *Seal finger-Erysipelothrix*

Burns

- *P. aeruginosa*
- *S. aureus*
- Enterobacteriaceae

- **Bites** – *Pasteurella, Eikenella, Capnocytophaga*
- **Cat scratches** – *Bartonella*

Opportunistic Infections

- Anaerobes-non sporing
- *S. aureus*
- *P. aeruginosa*
- Enterobacteriaceae

Pseudomonas aeruginosa SKIN INFECTIONS

can grow in disinfectants - why so present in hospital infections

Burns infection

The most common cause of burns infections is Pseudomonas.

Wound infections especially Diabetic foot infections. Colonization of shoes/ contamination of wound from environmental source

Pseudomonas aeruginosa SKIN INFECTIONS

Hot tub Folliculitis

Community acquired infection following exposure to contained contaminated water (whirlpools, swimming pools, water slides, bathtubs), use of diving suits etc.

Predisposing conditions:
IMMUNOSUPPRESSED host
CHEMOTHERAPY
NEUTROPENIC

Ecthyma gangrenosum

- Infection of the skin often seen in immuno-compromised patients such as those with neutropenia
- Lesion: round/ oval with a halo of erythema and necrotic center
- Single or multiple and heal with scar formation.
- Invasion of blood vessels by Ps. aeruginosa causing infarction and necrosis. (Exotoxin A)

small papule with black center

Pseudomonas aeruginosa

- Gram negative bacilli, aerobic
- Oxidase positive
- Fruity odor (grape –like)
- Pigment production-pyocyanin (fluorescein), pyoverdin etc.

Virulence factors

Enzymes: Elastases, Phospholipase C.

Toxin: Exotoxin A – acts by inhibiting protein synthesis by ADP ribosylation of EF-2.

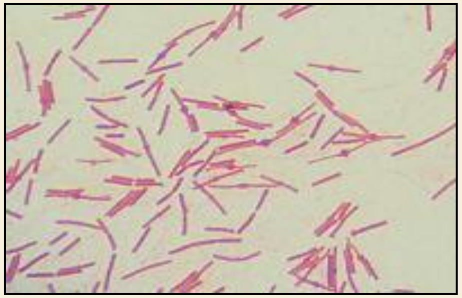
Treatment

C. diphtheria similar

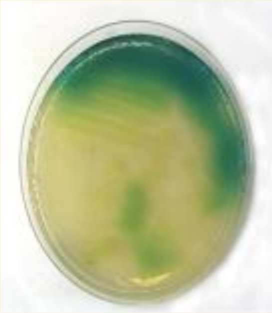
good to use combo therapy

- Piperacillin + aminoglycoside
- Multidrug resistant strains emerging.
- **Prevention**- avoid flowers in hospitals. Avoid moisture and environmental source contamination.

can also thrive in liquid soap



Gram Negative Bacilli

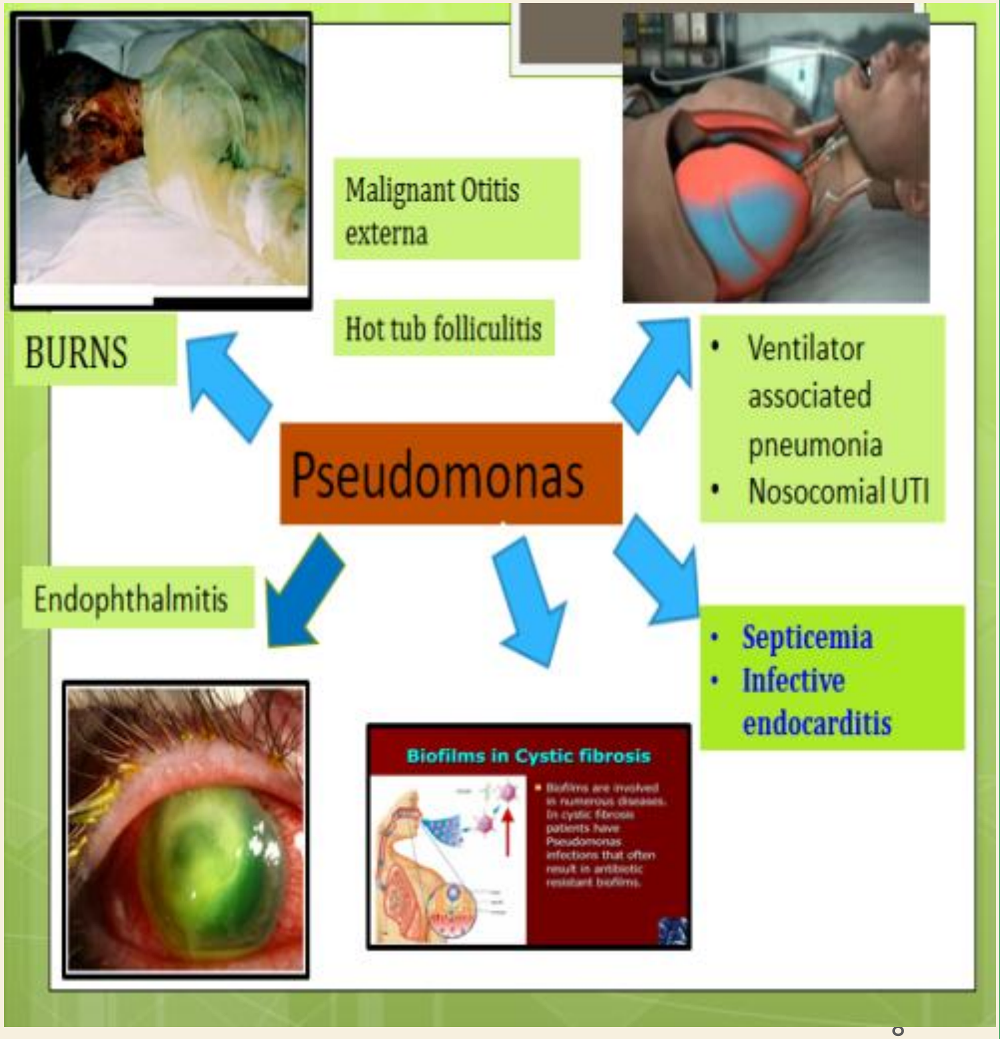


Nutrient agar showing green pigmented colonies

Mac Conkey's agar showing non lactose fermenting (NLF) pale colonies

PSEUDOMONAS AERUGINOSA INFECTIONS (THE BIG PICTURE)

- Opportunistic pathogen causing:
  - **P** = pneumonia, especially in cystic fibrosis patients and patients on ventilators
  - **S** = sepsis
  - **E** = external otitis (swimmer's ear)
  - **U** = urinary tract infections
  - **D** = drug use and diabetics
  - **O** = osteomyelitis
  - Burn and wound infections
    - Nosocomial infections
    - Hot tub folliculitis



OBLIGATE ANAEROBES

CLOSTRIDIA

BACTEROIDES

FUSOBACTERIA

PEPTOCOCCUS

PEPTOSTREPTOCOCCUS

missing superoxide dismutases and catalas

Gas Gangrene
Bacterial/ Clostridial Myonecrosis

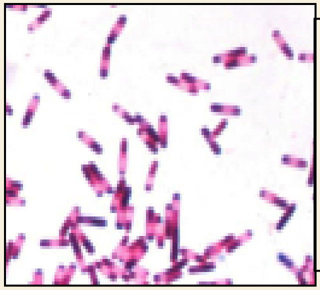
- Clostridial myonecrosis, or **gas gangrene**, is a fast-spreading and potentially life-threatening necrotic **infection caused due to the multiplication of bacteria (mainly Clostridium spp).**
- **Causative agent- multifactorial, in which C. *perfringens* plays the predominant role.**

•**Gram positive bacilli**, with sub-terminal spores that are resistant to heat, drying and several chemicals.(Sensitive to Glutaraldehyde, formaldehyde and Ethylene oxide)

•Non-motile, obligate anaerobic

•Form several toxins and enzymes.

•12 different toxins formed out of which an important toxin is Alpha toxin/ Phospholipase C (lecithinase)- detected by Nagler's reaction/ Immunoassays



Gram stain of culture smear of *C.perfringens* showing subterminal spores

- Anaerobic infections are associated with foul smelling pus.
- They lack enzymes- Superoxide dismutase and catalase hence die in the presence of O_2

PATHOGENESIS AND CLINICAL FEATURES OF GAS GANGRENE

Predisposing factors

- Road Traffic injuries, Puncture wounds, Gunshot wounds, War injuries, Compound fractures.

Rarely:

- Following surgeries like amputation for peripheral vascular diseases .
- Following injection of Adrenaline.

Type A Cl. perfringens inhabits intestinal tracts of humans and animals and is widely distributed in nature , esp. in soil and water contaminated with feces.

↓

Introduction of Clostridial spores by trauma or surgery

↓

Low redox potential in damaged tissues creates anaerobic environment, spores germinate and vegetative forms multiply

↓

Produce toxins: Lecithinase (α-toxin), θ toxin, collagenases, hyaluronidases etc.

↓

Extensive damage to tissues, cell membranes, RBCs, leukocytes, platelets with myonecrosis, cellulitis , increase in vascular permeability and extravasation.

- Extensive damage to tissues cell membranes and muscles.
- Abundant gas formation/ crepitus can be felt. This further reduces the blood supply, extending the area of anoxic damage.
- Increase in vascular permeability and extravasation can lead to shock, renal failure and death.

Laboratory diagnosis
Treatment/ Prevention

Sample: Tissue biopsy/ deep wound swabs.

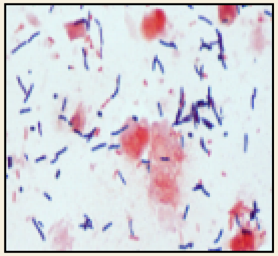
Microscopy- Grams stain of wound swab- Scanty inflammatory cells and GPB.

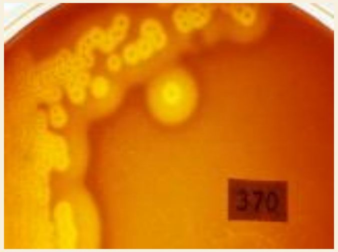
spores not seen in tissues

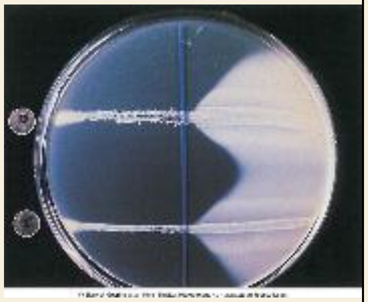
Culture: Blood agar (anaerobically incubated): Double zone hemolysis.

- Stormy fermentation in milk media:-acid and gas formation in Litmus milk medium
- Confirmatory tests: Nagler's reaction on egg yolk agar

neutralization test with anti-toxin







Neutralization test for the detection of activity of α-toxin also called Lecithinase or Phospholipase C.
Demonstrated on Egg Yolk Agar
Lecithinase (α-toxin; phospholipase)released by the clostridial strain, hydrolyzes phospholipids in egg-yolk agar around streak on right(seen as a pearly opacity).
Antibody against α-toxin inhibits activity around left streak due to neutralization of the toxin.

Treatment

- Aggressive treatment with surgical debridement /amputation if needed.
- High –dose Penicillin therapy OR Penicillin+ Clindamycin combination.
- Hyperbaric oxygen therapy

Prevention

- Good wound care
- Use of prophylactic antibiotics

NON-SPORE-FORMING ANAEROBES

GPR

Actinomyces spp.

Propionibacterium acnes

GNR

Bacteroides spp.

Fusobacterium spp.

Prevotella melaninogenica

GPC

Peptococcus

Peptostreptococcus

NON-SPORING ANAEROBIC ORGANISMS

Part of normal flora; Opportunistic pathogens

Infection occurs when host defenses and anatomical barriers are altered (tissue ischemia, trauma, surgery, perforation of intestine…)

Often mixed infections: the growth of facultative anaerobes facilitates obligate anaerobic growth

- *Bacteroides fragilis*
  - Gram negative bacilli
  - Predominant gut anaerobe
  - Intra-abdominal and soft tissue infections below the waist
    - Surgical wounds
    - Decubitus ulcers
  - Septicemia
  - Antibiotic resistance is common

- *Prevotella melaninogenica & Fusobacterium*
  - Gram negative bacilli
  - *Prevotella* – GI & oral flora
  - *Fusobacterium* – GI, oral, upper RT flora
  - Infections above the waist
    - Sinuses, ear and periodontal infections
    - Brain abscesses
    - Aspiration pneumonia (alcoholics)

Anaerobic Gram-Positive Cocci

Peptococcus and *Peptostreptococcus*

Brain abscesses, sinusitis, intraabdominal infections, skin and soft tissue infections

Aspiration pneumonia

Diagnosis- Growth under anaerobic conditions, biochemical testing and gas chromatography

Treatment

Metronidazole

Most produce beta-lactamases

Surgical drainage

Klebsiella and mycobacterium tb

Necrotizing fasciitis
Most common agent- *Strep pyogenes*

- A severe, destructive lesion involving the fascia and destruction of underlying muscle and fat. Caused by **flesh eating bacteria-**may be mixed/ **usually due to Group A Strep (GAS).**
- Can cause death due to multiple organ failure.
- Rapid spreading growth of streptococci and action of SPE toxin.
- Differential diagnosis- Gas Gangrene

bacterial tonsilitis

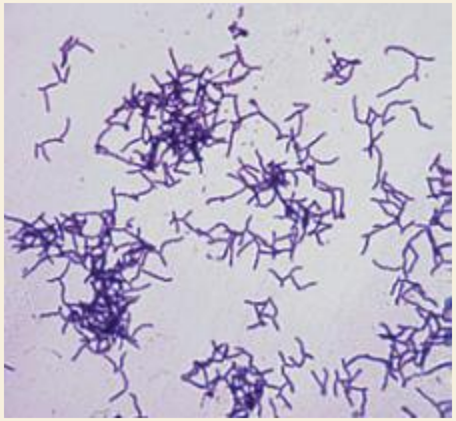
super antigen = non specific t lymphocytes and capable to stimulating other t cells so causes cytokine storm

Erysipelothrix rhusiopathiae

Erysipeloid/ Whale finger/ Seal finger

- Violaceous lesions caused by *Erysipelothrix rhusiopathiae*
- Wounds caused due to fishing / handling whales/ seals/ pigs.(Occupational disease)
- **Gram positive bacilli** that are catalase negative.
- Treatment: Penicillin

resistant to vancomycin



may resemble c. diphtheriae

Vibrio vulnificus

Found in salt water; may contaminate oysters

Cuts when opening oysters

Wound infection ➡ invasion ➡ necrotic cellulitis

- Virulence factors
  - ➢ Capsule, cytotoxic enzymes, LPS

liver disease or hep very susceptible to sepsis

- Diagnosis
  - ➢ Microscopy – Gram negative curved bacilli
  - ➢ Culture
    - Halophilic: requires salt for growth
    - Thiosulfate Citrate Bile Salts Sucrose agar (TCBS) – selective and differential alkaline media- usually greenish colonies.
  - ➢ Identification
    - Oxidase positive, ferments lactose (unlike other vibrios)
- Treatment – Doxycycline + 3rd Gen Cephalosporins

WOUND INFECTIONS
ANIMAL BITES/ SCRATCHES

- **Bites** – *Pasteurella, Eikenella, Capnocytophaga*
- **Cat scratches** – *Bartonella*

Bartonella henselae causes

CAT SCRATCH DISEASE

Gram –negative rods which do not stain well with gram's stain

Better stained with silver stains such as Warthin-Starry stains.

Usually do not grow in culture

BARTONELLA-
CLINICAL PRESENTATION/ DIAGNOSIS/ TREATMENT

Immuno-competent host:

✓Benign disease manifesting as local papule and regional lymphadenopathy.
✓Entry of organism into regional lymph nodes via the bloodstream causes chronic regional lymphadenopathy.
✓A granulomatous host reaction.
✓Bacteria may be seen in lymph nodes but are very scanty.

HIV

Immuno-compromised host:

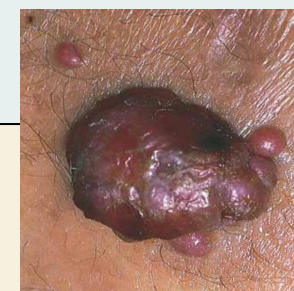
✓Severe spectrum of diseases.
✓Bite results in a local papule followed by extensive multiplication of Bartonella with bacteremia and involvement of lymph nodes, skin , liver and spleen.
✓Fevers with bacillary angiomatosis (vascular proliferative disease of the skin) manifesting as blood filled nodules and bacillary peliosis (hepatitis).

similar to Kaposi sarcoma but is diff

Hand showing local papules at the site of bite

Severe axillary lymphadenopathy

Diagnosis:
✓Silver stain/Serological tests/ PCR
✓Cultures are usually sterile
Treatment: Azithromycin / Doxycycline



Skin lesion of bacillary angiomatosis in AIDS patient with *B.henselae* infection

Pasteurella multocida- associated with bites of cats/ dogs.

Gram negative rods/ cocco-bacilli.

Non-motile

Facultative anaerobe

Grow on Chocolate and Blood Agar

No growth/ poor growth on MacConkey's agar

Oxidase +

Virulence factors: Capsule; Endotoxin

Source of infection-Oral cavity / upper respiratory tract of cats and dogs.
Transmission- scratch/ bite/ salivary contamination
Wound infection
Cellulitis
Lymphadenitis
Pulmonary dysfunction in immuno-compromised.
Sometimes, Septic arthritis/Osteomyelitis

- ✓Treatment
- ✓Thorough cleaning of wounds
- ✓Amoxycillin + Clavulanic acid
- ✓Macrolides are usually not given(due to resistance)

OTHER BITE- ASSOCIATED INFECTIONS

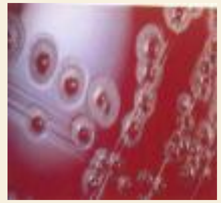
HUMAN BITE

- *Eikenella corrodens* part of oral flora

  Part of HACEK group — cardiovascular infections

  infective endocarditis

  – Corrodes agar
  – Bleach-like odor
  – Human bites or fist fight injuries
  – Cellulitis



DOG BITE

- *Capnocytophaga canimorsus*
  - GNR
  - Dog bites
  - Cellulitis


Thank
you!!